Microeconomics 2
The Consumer Problem
Instructor Notes

Arseniy Stolyarov

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Live route for Lecture 2

Retain the approved opening and both KKT proofs. After KKT, establish Walras' law and convexity/uniqueness, then solve four models together: Cobb–Douglas, substitutes, linear plus logarithm, and complements. Only after the examples introduce indirect utility and its properties. State and prove a general constrained smooth-branch envelope theorem; apply it to prices and income. Then prove Roy using the fixed-bundle argument and recover demand for all four model families. Duality and Slutsky remain Lecture 3 reserve material.

Envelope regularity and Roy

The general envelope theorem assumes C1 problem data and a C1 branch of optimal choices and KKT multipliers on an open parameter neighbourhood. Keep all chain-rule terms before using stationarity, feasibility, complementary slackness and nonnegativity of differentiable multiplier branches. This is a sufficient smooth version. Roy has a separate direct proof requiring continuous LNS utility, existence, differentiable indirect value and a positive wealth derivative; it does not require a smooth direct index or differentiable demand.

Conventions and sources

Use $p \gg 0$, $m > 0$, D for demand sets, x for a unique vector and v for the value. The quasilinear example is now $u = a + 4\log(1+q)$ with $a + rq \leq m$, so the unit-price target is q=3. The old square-root example has been replaced. Saddle-domain qualifications and corrected source p.164 price/index typos are retained. Inputs are the supplied Lectures 10–11. Homework 1 is due 14 November 2026.

Slide-by-slide notes

1. Microeconomics: Lecture 2 — The consumer problem

Briefly identify this as the next block, without repeating the entire course administration. Homework 1 is now issued after lecture block 3, not after this block.

2. From preferences to an actual choice

Three visible bullet points only. Follow the revised order: budgets and existence, saddle/KKT and both proofs, then structural demand tools, examples, envelopes and Roy. No board-stage directions are shown to students.

3. The picture we already know

Show three utility levels (2, 4, 6) on the second reveal, including an unattainable higher curve. The third reveal adds the optimum. The budget remains p=(1,1), m=8.

4. The consumer problem

Show the maximisation objective first, then the budget inequality and all quantity nonnegativity bounds underneath it. The parameter and feasible-set definitions follow.

5. What did we assume when we wrote that?

Use the spaced bullet list; the subscription/quantity-discount/indivisibility question is a final separate reveal.

6. Currency invariance of the budget

The budget currency-invariance claim appears first, then its proof. This is not yet a statement about utility or a unique demand function.

7. The budget is nonempty and compact

The compactness claim appears before the proof. Reveal nonemptiness, closedness and boundedness separately. Strictly positive prices supply finite coordinate bounds.

8. Does an optimum exist?

No concavity or monotonicity is needed for this existence statement. Keep existence, uniqueness and computation separate throughout. The term compact means closed and bounded here because the space is finite-dimensional.

9. Which assumption is doing the work?

For the last question use u=x1+x2 at p=(1,1), any m>0: every budget-line bundle is optimal. The zero-price example is a failure example, not an assertion that all zero-price problems lack solutions.

10. Marshallian demand is initially a correspondence

Uppercase D is the correspondence and lowercase x the unique vector. This avoids using x(p,m) sometimes as a set and sometimes as a vector without warning. Marshallian demand is also called uncompensated demand.

11. Proof: demand is homogeneous of degree zero

The result is equality of correspondences, so no uniqueness assumption is necessary. A structural demand model should respect this restriction when all prices and nominal wealth are rescaled together.

12. Let a multiplier enforce the budget

Use a separate symbol ell for the budget-only Lagrangian. Its x-domain remains the nonnegative orthant, so utility is never silently evaluated at negative consumption. The full L with mu appears later for the first-order system.

13. Why minimise in the multiplier?

The inner expression is an infimum, since infeasible bundles send the value to minus infinity. It is an exact extended-real encoding of the budget, not a minimax interchange theorem.

14. A global saddle point

This is now BEFORE KKT. Domain: x in the orthant, lambda nonnegative. The two inequality directions correspond to maximisation in quantities and minimisation in the budget multiplier. No claim that every arbitrary nonconcave optimum admits a saddle is being made.

15. Proof: the saddle enforces feasibility and slackness

Feasibility of quantities is imposed by the saddle domain, not derived by varying missing bound multipliers. Budget feasibility and its complementary slackness follow directly from minimisation over lambda.

16. Proof: a global saddle gives an optimal bundle

The three-inequality chain proves global optimality directly. Concavity is unnecessary for this direction; it matters when trying to establish the global

maximisation inequality from local derivative conditions.

17. How could we find or verify a saddle?

Bridge to the board. Under concavity, KKT will verify the budget-only saddle; the appendix contains the short supporting-plane proof. Do not present arbitrary stationarity or a minimax interchange as sufficient.

18. Why compare marginal utility per unit of money?

The feasible direction requires enough of good j to reduce it; this motivates why equality applies to consumed goods, while goods at zero require an inequality. The derivative calculation is local, not yet a global optimality proof.

19. The Lagrangian: one multiplier per inequality

The plus signs match the maximisation convention with constraints written as nonnegative. Do not mix with the minimisation convention. A log utility is not defined at zero: for that example establish an interior optimum first and work on the positive domain.

20. Stationarity in quantities

The notation u_i is partial derivative with respect to x_i. Because quantity bounds are represented explicitly, a corner does not mean deleting stationarity: its bound multiplier enters the equation.

21. The four Kuhn–Tucker groups

For p strictly positive and m>0, the active linear constraints satisfy the necessary constraint qualification. Differentiability on a neighbourhood is required for this form. Necessity does not require concavity; global sufficiency does require additional structure.

22. Complementary slackness: read the products

No. A binding constraint may have zero multiplier. In particular, a budget can bind at a point where marginal value is zero. Appendix zero-shadow-value gives an explicit continuous LNS example. For substitutes at equal returns, an active zero-consumption bound may also have zero multiplier.

23. For consumed goods: equal marginal returns

Define MRS as u_i/u_j only where the denominator is positive and the indifference curve is differentiable. If all relevant marginal utilities vanish, division is invalid. Corners and kinks are handled differently.

24. For a good at zero: an inequality

It would force purchases of goods with too low a marginal return or generate negative quantities. The inequality is meaningful with positive prices. It is a necessary marginal condition, and under concavity part of a globally sufficient system.

25. A candidate is not yet a solution

The source has an interior-looking theorem followed by a broader constraint-qualification footnote. We state the smooth necessary result at corners too, with the qualification explained in the appendix. Concavity is a sufficient condition for KKT sufficiency, not a necessary property of preferences.

26. Theorem 1: KKT is necessary

This is the first explicit theorem on source Lecture 11 p.171, strengthened to include corners. The direct constructive proof supplies regularity for this particular linear budget; concavity is not used. We assume smoothness on a neighbourhood so gradients at the bounds make sense. Necessity holds even for a local optimum.

27. Construct the budget multiplier from an optimum

This is the construction in source Lecture 10 pp.162–164, corrected. Work at a nonzero optimum. Equal marginal returns for consumed goods follow from the preceding feasible-trade argument; nonnegativity follows by slightly reducing a consumed good. For the excluded zero-bundle case: if $x^* = 0$ is optimal and $m > 0$, a small increase in each coordinate is affordable, hence $u_i(0) \leq 0$. Set $\lambda^* = 0$ and $\mu_i^* = -u_i(0) \geq 0$; stationarity and complementary slackness hold. No concavity is needed for the sign argument.

28. Proof 1: consumed goods have one common return

For a pair of positive quantities, either sign of a sufficiently small spending transfer is feasible and preserves cost. If normalised marginal utilities differ, the transfer toward the larger one has positive directional derivative, contradicting even local optimality. This completes the multiplier-independence argument invoked on the construction slide.

29. Construct the quantity-bound multipliers

The equality $\mu_i^* = \lambda^* p_i - u_i(x^*)$ is the definition chosen to make stationarity true, not an implication that assumes dual feasibility. For consumed goods set mu to zero; the common-return result makes stationarity hold there too. The next two slides prove the missing sign.

30. If a constructed multiplier were negative

This is plain algebra and division by $p_i > 0$. In the source p.164 the left normalised derivative is mistyped: the consumed-good term must be u_j/p_j, not u_i/p_j. The same page later omits the price denominator when writing lambda for a consumed good. Both formulas here are corrected.

31. Proof: a profitable spending transfer

The spending transfer is one-sided because the unconsumed good can be increased and the positive holding of j can be reduced. Positive directional derivative implies a strict improvement for all sufficiently small positive epsilon by differentiability; the finite utility change is not asserted to equal its linear approximation. This proves only $\mu_i \geq 0$, not strict positivity, including at a zero quantity.

32. Proof 1: finish with complementary slackness

This is the missing end of the source p.164 proof, completed on pp.165 and 169. Under budget slack a consumed coordinate can move in both directions, so its derivative is zero and lambda=0. Bound products are zero by construction, not by assuming the conclusion. The source p.165 has a reversed statement about which quantities make mu zero; the correct rule is positive quantity implies zero multiplier.

33. Proof 1: the zero-bundle case

This gives the separate source p.162 footnote in the main proof. Small positive coordinate increases are affordable since m>0. Their derivatives cannot be positive at a local optimum. Zero budget multiplier and minus the utility gradient supply all KKT conditions.

34. Theorem 2: concave KKT is sufficient

This is the second theorem on source Lecture 11 p.171. Concavity of u on the nonnegative orthant and differentiability near the candidate are sufficient. The source footnote gives a weakening using quasiconcavity PLUS a strict gradient implication. We use the simpler concavity theorem here. Quasiconcavity alone does not suffice, as the existing counterexample shows.

35. Fix the multipliers and look only at quantities

Fix the multipliers before discussing curvature in quantities. Affine terms preserve concavity, and constructed stationarity makes the gradient zero. Evaluate on a convex domain where u is defined and concave, for example the consumption orthant with a differentiable extension near the candidate. Do not silently extend log utility to negative quantities. The original source assertion for all real vectors needs a utility domain supporting it.

36. Proof: a stationary point of a concave function

The supporting-hyperplane inequality proves a global maximum, not just a local one. Concavity plus zero gradient is sufficient; strict concavity is needed only as a sufficient condition for uniqueness. A stationary point of a generic nonconcave function need not be optimal. The statement concerns the Lagrangian at fixed multipliers; feasibility and complementary slackness are still needed to connect it to consumer optimality.

37. A horizontal tangent above the whole graph

The first reveal shows the function, the second its tangent and candidate, and the third explains the word flat in the source. This is a one-dimensional illustration of the supporting-plane argument; the preceding slide is the multidimensional proof.

38. Proof 2: return from the Lagrangian to utility

The three inequalities prove consumer optimality from KKT without omitting feasibility or slackness. For the saddle interpretation we take the quantity domain to be the nonnegative orthant, where u is defined and concave. Full-L saddle minimisation follows from nonnegative slacks and zero products; no claim for negative quantities outside the domain is made.

39. Budget exhaustion: Walras' law

This is the consumer budget-exhaustion property often called Walras law. It is not a proof of existence. Saving is not ruled out if it is one of the goods. See the first lecture for satiation counterexamples.

40. Proof: LNS and budget exhaustion

The strict radius handles feasibility cleanly. Only an existing optimum is being considered. This proof is shared conceptually with lecture 1.

41. Proof: convex preferences give a convex demand set

This is an ordinal result using convex preferences, not concavity of the particular utility representation. It does not establish nonemptiness; the existence assumptions are maintained.

42. Strict convexity: at most one optimum

Answer to the last question: no. Strict convexity provides at most one optimiser on a convex feasible set. Continuity plus compactness supplied existence separately. Do not confuse strictly convex preferences with a strictly convex utility function.

43. Proof: uniqueness from strict convexity

No strict concavity of the chosen utility representation is required. For Cobb-Douglas on the closed orthant, establish positive optima before invoking strict properties on the positive orthant.

44. Before differentiating, inspect the economics

The source handouts emphasise not using the Lagrangian as a cookbook. Keep the method appropriate to the functional form. A nondifferentiable kink is not fixed by pretending that a derivative exists.

45. Checkpoint

Answers: continuity on nonempty compact budget; strict convex preferences plus existence; nonnegative bound multipliers and slackness; concavity is a sufficient condition; a corner, infeasibility, a stationary nonmaximum or a nondifferentiable kink can be missed. No timed exercise label is used.

46. Example 1: Cobb–Douglas

Ask students about boundary and interior candidates before revealing the positivity argument; logs are used only after ruling out boundary optima.

47. Choose a convenient representation

The log transform preserves the positive-domain ranking. Do not apply the compactness theorem directly to the open positive budget; the previous raw-utility argument already supplied existence and excluded zero coordinates.

48. Solve the interior conditions

Concavity of the log objective and feasibility establish sufficiency. Positivity is satisfied because alpha, 1-alpha, prices and wealth are positive. The first-order equations are not being used alone without an optimality justification.

49. Interpret the solution

Constant expenditure shares and their invariance to prices/wealth are restrictions specific to Cobb-Douglas. Exhaustion here follows from preferences, while homogeneity follows from the budget construction. Do not present Cobb-Douglas shares as a fact about all consumers.

50. A price counterfactual

New x=(2,6), spending remains (6,6). Raw utility goes from sqrt(18) to sqrt(12). There is no meaningful additive utility contribution for each good in the

multiplicative raw representation. The question is for students; the slide does not reveal the answers.

51. Example 2: perfect substitutes

Ask about spending returns and multiplicity before revealing the three regimes.

52. Verify a corner with the full KKT system

All KKT groups hold, and the objective is affine/concave, so they certify global optimality. At the corner the subjective ratio 2 is not equal to the price ratio 3, and it should not be.

53. Equal returns: a whole set of demand

Indirect utility is not yet introduced: describe the segment of optimal bundles and their common utility value without v notation.

54. Example 3: lessons and money left over

The new model is linear plus a shifted logarithm: $u = a + 4\log(1+q)$, $a + rq \leq m$. Price a is normalised to one. The shift allows zero consumption and makes direct utility smooth there.

55. Solve first, then respect the quantity bounds

The reduced objective is strictly concave; clip its stationary point to the feasible interval. This proves uniqueness even though the original index is affine in a. Unit-price target is q=3, replacing the old square-root example with target 4.

56. Three regimes, one decision problem

At r>=4: lambda=1, mu_q=r-4. At r<4 and m<=4-r: lambda=4/(r+m), mu_a=lambda-1, mu_q=0. In the interior lambda=1 and both bound multipliers vanish. At m=4-r the formulas coincide. At r=1, m=2 gives (0,2), m=8 gives (5,3).

57. Example 4: perfect complements

Seek a useful-proportions or target-cost argument first. Smooth tangency at the kink is invalid. This is a direct utility-maximisation example, not the later full Hicksian programme.

58. A fixed-proportion solution

The cheapest implementation of the highest attainable target exhausts wealth; any extra quantity would exceed it. Positive prices give uniqueness.

59. A practical solution checklist

This is the main procedural summary for exercises. A formula with a negative quantity is not an acceptable answer to the nonnegative consumer problem. A smooth method is not a reason to discard nonsmooth optima.

60. From a classroom solution to a structural model

This connects to the lecturer’s requested motivation about real structural models. Identification of alpha is conditional on the maintained Cobb-Douglas model and complete measurement of the modelled budget. It is not an unconditional causal inference from one purchase.

61. Indirect utility: the optimal value

v is a function even if Marshallian demand is a correspondence. It is ordinally scaled: changing the representation u changes v by the corresponding increasing transform.

62. Indirect utility: the properties to prove

The price statement is weak, not necessarily strict for a single price. LNS gives strict increase in wealth, but a derivative can still be zero at a point. Continuity follows from the maximum theorem; it is not differentiability.

63. Proof: currency units and higher prices

Budget inclusions are the whole argument. Hold the numerical representation u fixed. The price comparison is weak, and the vector inequality means all prices weakly increase.

64. Proof: more wealth and local nonsatiation

Use the same epsilon-ball argument as budget exhaustion, now spending the new slack to obtain a strictly better bundle. LNS supplies strictness. This does not assert a strictly positive derivative everywhere.

65. Continuity: put every budget on the same simplex

The expenditure-share simplex is a common compact set for every positive budget. Do not require exhaustion: the sum is at most one, not exactly one. No monotonicity, LNS or uniqueness is needed.

66. Proof: the optimal value is continuous

The maximum difference is bounded by the uniform difference of the objectives. Uniform continuity follows on a compact positive parameter neighbourhood times

the simplex. Finite early sequence terms are irrelevant. This gives a complete short maximum-theorem argument in this particular budget problem.

67. A less obvious property: quasiconvexity

This is joint quasiconvexity in prices and wealth, distinct from quasiconcavity of direct utility in consumption. The source proof has priming typos; the same candidate bundle x must be used at both endpoint budgets.

68. Proof: indirect utility is quasiconvex

The identical bundle x must occur in both endpoint inequalities. The proof does not require convex preferences, concavity or LNS, only well-defined optimal values on the relevant budgets.

69. Compute indirect utility in the same example

This v is the value of log utility, not the raw Cobb-Douglas value. Never combine its derivatives with the raw multiplier without accounting for the change of scale. The raw value is exp(v).

70. A continuous value need not be differentiable

Left derivative at p1=1 is -8, right derivative is zero. v is continuous but not differentiable. This example motivates the differentiability hypothesis in Roy's identity. It is not a failure of existence.

71. The envelope question

Present the envelope idea before derivative formulas. Do not mechanically differentiate a chosen x(p,m) and forget the binding constraints. The direct Lagrangian envelope includes the multiplier adjustment required by feasibility.

72. A general parameterised optimisation problem

All data are C1 on an open ambient decision/parameter domain. Quantity bounds are explicit inequalities. Parameters can enter the objective and every constraint. Equality multipliers have no sign restriction.

73. The general Lagrangian

A precise sufficient version assumes a C1 selection of optimal decisions and KKT multipliers on an open parameter neighbourhood. These assumptions need verification; they are not consequences merely of value differentiability. This is a smooth-branch theorem, not a claim covering every corner or kink.

74. Envelope theorem: a constrained smooth version

The general theorem is componentwise in a vector parameter. On the right, differentiate in the parameter holding quantities and multipliers fixed, and then evaluate at the optimum.

75. Proof: differentiate along an optimal branch

Identify V with L along the whole optimal branch using feasibility and complementary slackness. Keep the choice and multiplier derivative terms when differentiating; the next slide justifies their disappearance.

76. Proof: why multiplier derivatives disappear

For a positive inequality multiplier, its slack is zero. A zero multiplier on a differentiable nonnegative branch at an interior parameter has derivative zero. Either way g_j times lambda_j prime is zero. Equality slacks and quantity-stationarity terms vanish separately.

77. Apply the theorem to the consumer

Apply the general theorem only in regular smooth regions. In the consumer problem parameters enter the budget only. A binding budget alone need not imply a positive multiplier.

78. The budget multiplier has an interpretation

For Cobb-Douglas log utility lambda=1/m. For the raw degree-one representation lambda=v_raw/m. Both give the same bundle. Strictly increasing v can have a zero derivative at a point, so positivity in Roy cannot be omitted.

79. Roy’s identity: recover demand from value

The fixed-bundle proof needs continuous LNS utility, existence, differentiable v and v_m>0. It does not assume differentiability of u or demand. The smooth consumer-envelope ratio motivates it, but the separate proof covers a broader setting.

80. Roy without differentiating demand

This proof uses continuous LNS preferences, existence, a differentiable indirect value and a positive wealth derivative. It avoids assuming demand is differentiable or a priori single-valued, and does not require concave direct utility.

81. Check Roy in the Cobb–Douglas example

Use the same log utility scale throughout this calculation. This is an actual check against a separately derived demand, not only a restatement of the identity.

82. Roy example 2: substitutes and a corner

Differentiate the strictly active value branch. The price derivative for the unconsumed good is zero, correctly yielding zero demand.

83. The other corner and the tie

At equal spending returns the two active price gradients differ, hence v is nondifferentiable in prices and the Roy ratio is not available.

84. Roy example 3: the logarithmic model

Substitute all three demands into a+4 log(1+q). Values and first derivatives match at r=4 and m=4-r. This is no longer the square-root example.

85. One ratio, three regimes

At the two transition boundaries, the value is C1 even though demand has a kink. The smooth-choice-branch envelope proof may fail there, but the direct Roy proof still works. Recover q from the price derivative and a by budget exhaustion with its unit price.

86. Roy example 4: a kink in direct utility

The min index has a kink at the optimum; v=m/(p1+p2) is smooth at positive prices and wealth. The direct Roy proof applies. Distinguish smoothness of the direct index, the indirect value and the demand vector.

87. Demand survives a change of utility scale

The positive derivative condition is stronger than merely strictly increasing f. For instance a shifted cubic is strictly increasing but flat at its centre. Ordinal invariance of the optimisation problem remains true even if a particular differential formula degenerates.

88. Checkpoint

Answers: continuous u and nonempty compact budget for existence, strict convex preferences for uniqueness given existence; several maximisers share the same maximum value; marginal utility per price is no greater than lambda; concavity is a sufficient condition together with all KKT groups; differentiability of v and v_m>0 under the maintained consumer setup. Answers stay in the instructor notes.

89. Read next, and keep the core route clear

Assigned reading is the textbook. Use the already prepared Homework 1 on Lectures 1–2, deadline 14 November 2026; no old source-course HW5/HW6 notices.

90. Next block: two ways to ask the consumer question

This starts the optional block-3 preview from the tail of the supplied Lecture 11. It is not required before finishing the second-block core.

91. Expenditure minimisation

Capital H distinguishes a set of minimisers from a unique Hicksian vector h. A target above the attainable utility range makes the problem infeasible; do not state unconditional existence.

92. Why an unbounded feasible set can still have a minimum

The full upper contour set need not be bounded. Truncation by the cost of one feasible benchmark is enough. Continuity here refers to the selected utility representation.

93. Hicksian demand: basic properties

Targets are attainable, u continuous, p positive. If a nonzero minimiser exceeds the target, a slight proportional shrink remains feasible by continuity and is cheaper. If the target is u(0), zero is feasible and uniquely cheapest. For targets below u(0), no-excess utility need not hold.

94. The expenditure function has a different curvature

Holding the utility target fixed is essential. Concavity in prices does not require concave utility or convex preferences. Continuity on the attainable target domain is discussed under continuous utility and LNS in the instructor notes; avoid unqualified endpoint or infeasible-target claims.

95. Fixed wealth versus fixed utility

The red budget holds wealth fixed. The dashed compensated budget holds raw utility at four. Verify the compensated cost: 2*(2sqrt2)+4sqrt2=8sqrt2. The three points use different markers, not colour alone.

96. Two identities to develop next time

This is a preview, not a full Slutsky derivation. At target u(0), the expenditure is zero and v(p,0)=u(0); interpreting this endpoint extends the value function beyond the

positive-wealth core. More generally the corresponding solution sets agree under the maintained assumptions.

97. A monetary compensation question

CV is stated with the sign convention of extra wealth required after a price increase. The old budget is exhausted so e(p0,u0)=m. Monetary units do not automatically remove interpersonal or distributional normative choices; this qualifies an overly broad claim in the source.

98. Why the constraint qualification holds here

This is LICQ for the original linear budget and bounds with p positive and m positive. It does not hold automatically in every nonlinear constrained optimisation problem. If m=0 and all bounds and budget bind, the argument fails.

99. Proof: why concavity makes KKT sufficient

If lambda is zero the budget term is zero; otherwise slackness gives p*x*=m and p*y<=m. For bounds, mu*x*=0 and mu*y>=0. This is a direct global proof and does not need a separate saddle-point theorem.

100. Proof: concave KKT gives the budget saddle

This proves that concave KKT yields a global saddle for ell(x,lambda) on x>=0, lambda>=0. Only the budget is relaxed in ell; mu encodes the first-order bound conditions. Do not assert global maximisation of the full L over negative quantities where the utility may be undefined.

101. A KKT point that is not optimal

This shows why quasiconcavity alone is insufficient; the source’s additional strict gradient condition is not assumed in this counterexample. Pseudoconcavity or other additional conditions can restore sufficient first-order tests, but they are not needed in this core lecture. The example also shows that KKT does not by itself impose budget exhaustion at every candidate.

102. A binding budget with zero marginal value

Replace the raw utility by x and its multiplier is positive. The same ranking can have a flat differentiable representation at the optimum. This is why Roy states v_m>0 rather than only strict increase of v.

103. Cobb–Douglas with any positive exponent sum

The product representation need not be concave when the exponent sum exceeds one, but the log representation is concave on the positive orthant and shares the

same optimum. The source formula has a price-index typo; the denominator uses the price of good i, not always p1.

104. Proof: expenditure minimisation has a solution

The attainable-target assumption is indispensable. No convexity or LNS is required for this existence argument.

105. Proof: homogeneity and price concavity of expenditure

Monotonicity in prices follows by evaluating the higher-price minimiser at the lower prices. No excess utility follows by a small proportional contraction of any nonzero minimiser with surplus utility; this also supplies strict target monotonicity above baseline and uniqueness under strictly convex preferences.